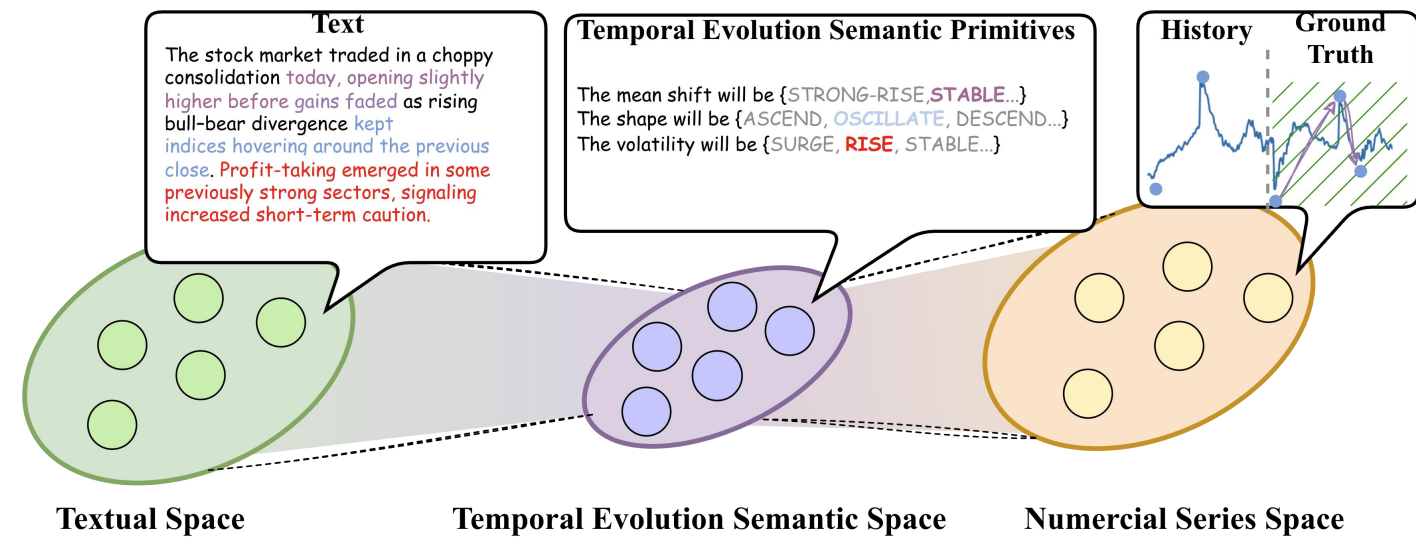


Outline

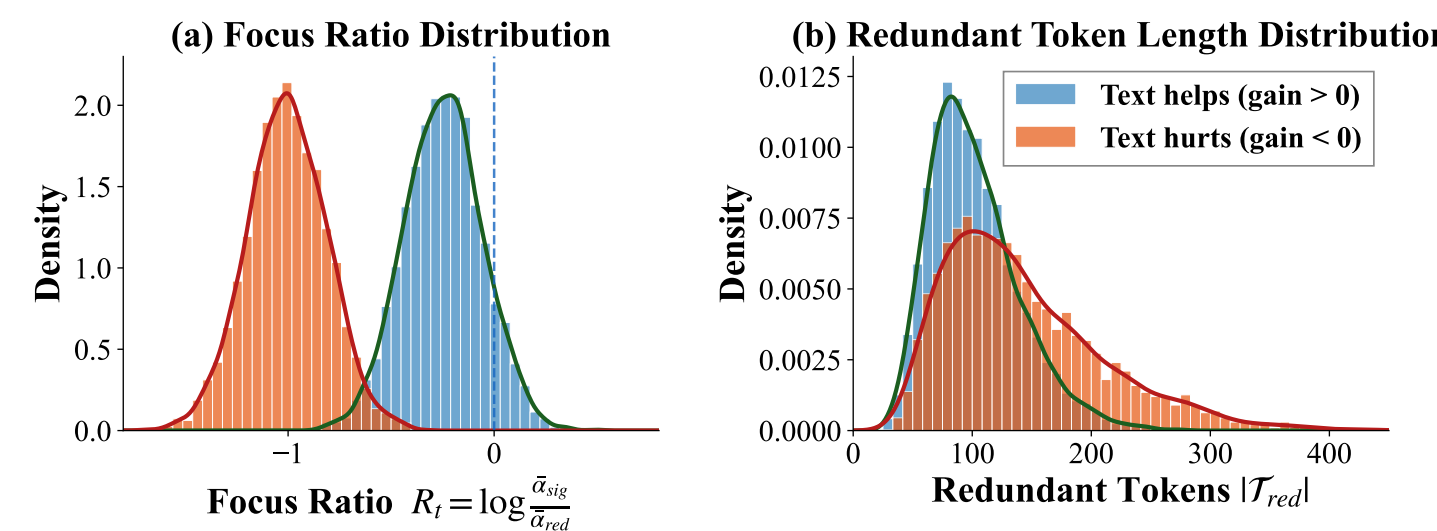
- **Problem:** exogenous events trigger *event-driven non-stationarity* that historical observations alone cannot anticipate.
- **Modality gap:** text expresses temporal impacts *implicitly and qualitatively*; forecasters rely on *explicit and quantitative* signals.
- **Approach:** TESS — a *Temporal Evolution Semantic Space* of numerically grounded primitives, extracted by an LLM and filtered through confidence-aware gating.
- **Up to 29% MSE reduction over state-of-the-art unimodal and multimodal baselines.**



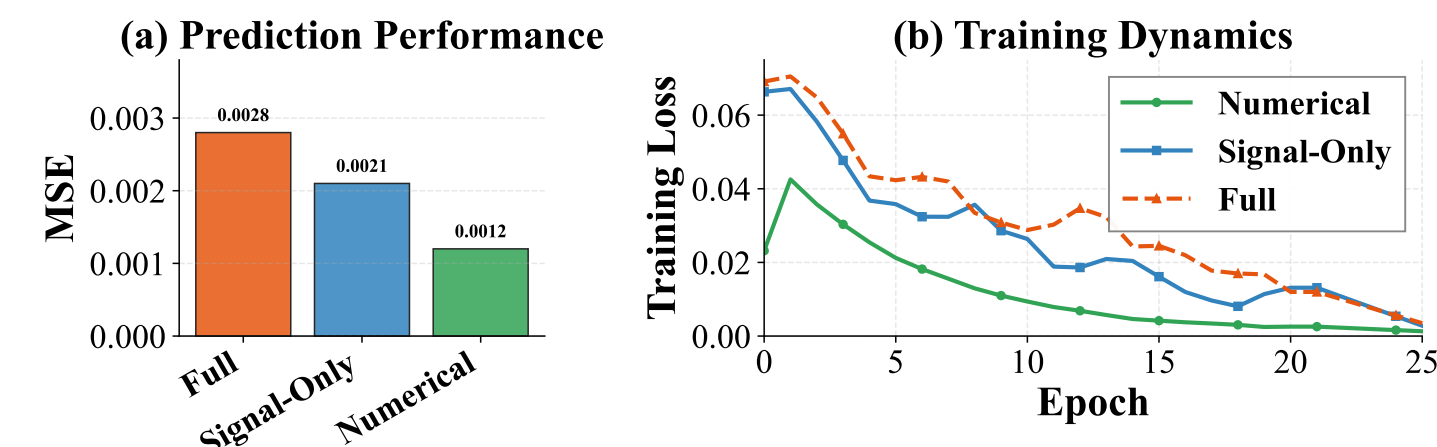
Cross-modal transformation via the Temporal Evolution Semantic Space.

Diagnosing the Modality Gap

Semi-synthetic benchmark: an LLM verbalizes future-window statistics (FNSPID), so the text provably contains predictive signals; tokens are auto-labeled into signal $\mathcal{T}_{\text{sig}}$ vs. redundant $\mathcal{T}_{\text{red}}$.

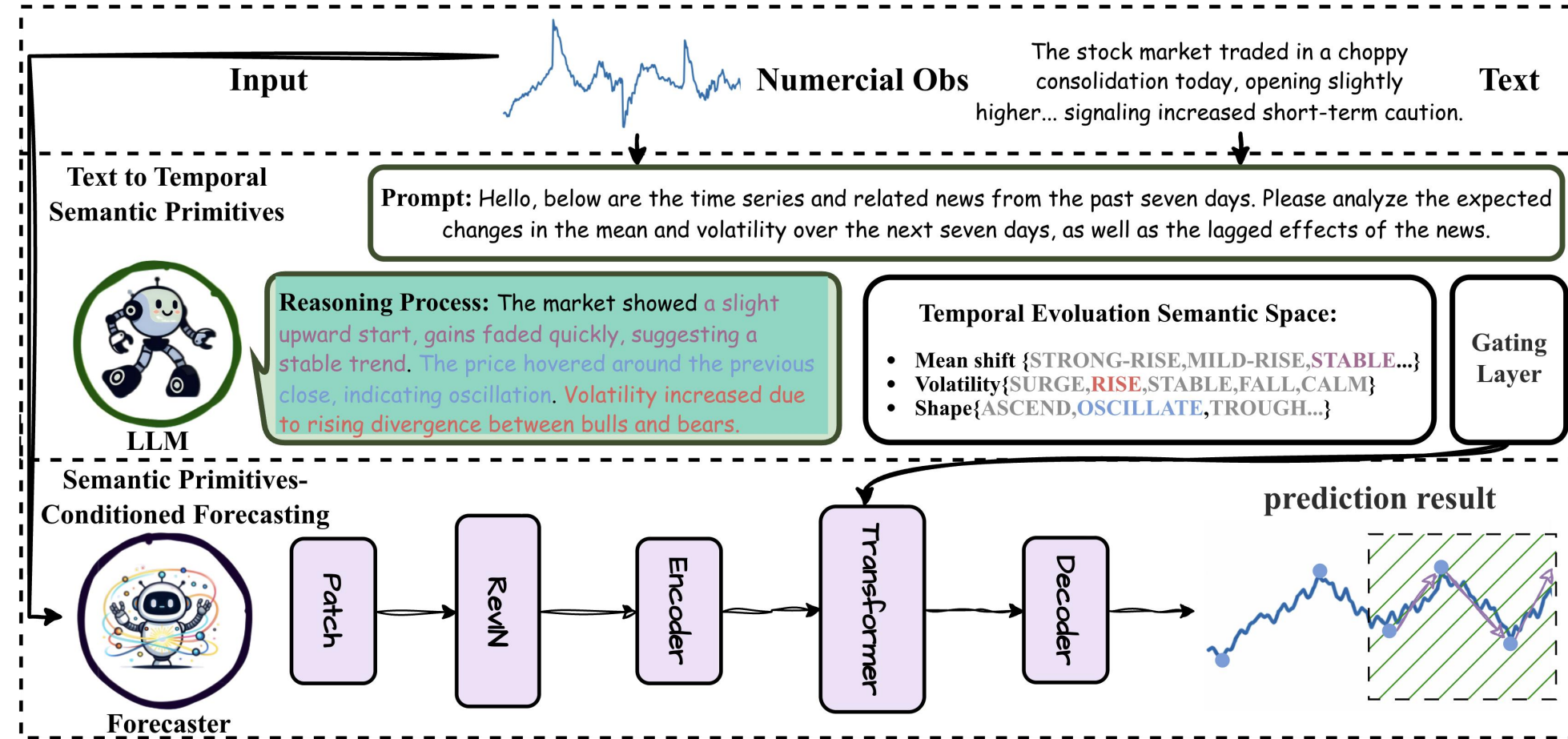


Finding ① — attention distraction. The vast majority of samples exhibit focus ratio $R_t < 0$: the model systematically over-attends to redundant tokens rather than predictive signals.



Finding ② — representational mismatch. Even with redundancy completely removed, Signal-Only remains inferior to Numerical — the modality gap persists.

Overall Architecture



A frozen LLM extracts primitives; gated primitives condition the forecaster.

① **Text → Semantic Space.** A frozen LLM classifies each primitive over $\mathcal{V}_k$ via structured prompting; **confidence-aware gating** maps the top-1/top-2 log-probability margin $m_{t,k}$ to a reliability weight:

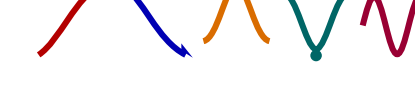
$$\ell_{t,k}(v) = \log P_{\text{LLM}}(v | s_t, \mathcal{D}_k), \quad g_{t,k} = \sigma(\mathbf{w}_k^\top [\mathbf{h}_{t,k}; \mathbf{W}_m m_{t,k}] + b_k).$$

② **Semantic Space → Numerical Space.** Gated embeddings $\tilde{\mathbf{h}}_{t,k} = g_{t,k} \mathbf{h}_{t,k}$ are stacked as *semantic prefix tokens* $\mathbf{P}$ and concatenated with PatchTST patch embeddings, conditioning attention at every layer:

$$\mathbf{Z}^{(0)} = [\mathbf{P}; \mathbf{E}_{\text{patch}}] \in \mathbb{R}^{(K+N) \times d}, \quad \mathcal{L} = \mathcal{L}_{\text{fctst}} + \lambda \mathcal{L}_{\text{gate}}.$$

Temporal Semantic Primitives

Each primitive's ground truth $\psi_k(\mathbf{Y}_t)$ is uniquely determined from the forecast window (**numerical verifiability**) — free supervision for gating.

Primitive	Vocabulary	Numerical grounding
Mean Shift Δ_μ		standardized difference $(\bar{Y}_t - \bar{X}_t) / \sigma(X_t)$
Volatility r_σ		$\log[(\sigma_Y + \epsilon) / (\sigma_X + \epsilon)]$ on differenced series
Shape		inter-patch trend signs $s_i = \text{sgn}_+(\tilde{u}_{i+1} - \tilde{u}_i)$
Lag & Decay		influence centroid c , tail mass d , peak prominence q

Theoretical Guarantees

- **Information-preserving restriction:** under semantic sufficiency, semantic encoders preserve $I(Z; Y | S)$ while not increasing $I(Z; X | S)$ (Thm. 1).
- **Generalization:** this restriction can only tighten the expected generalization bound (Thm. 2).
- **Soft-gating stability:** error attenuated $\propto g_{t,k}^2$ (Thm. 3).

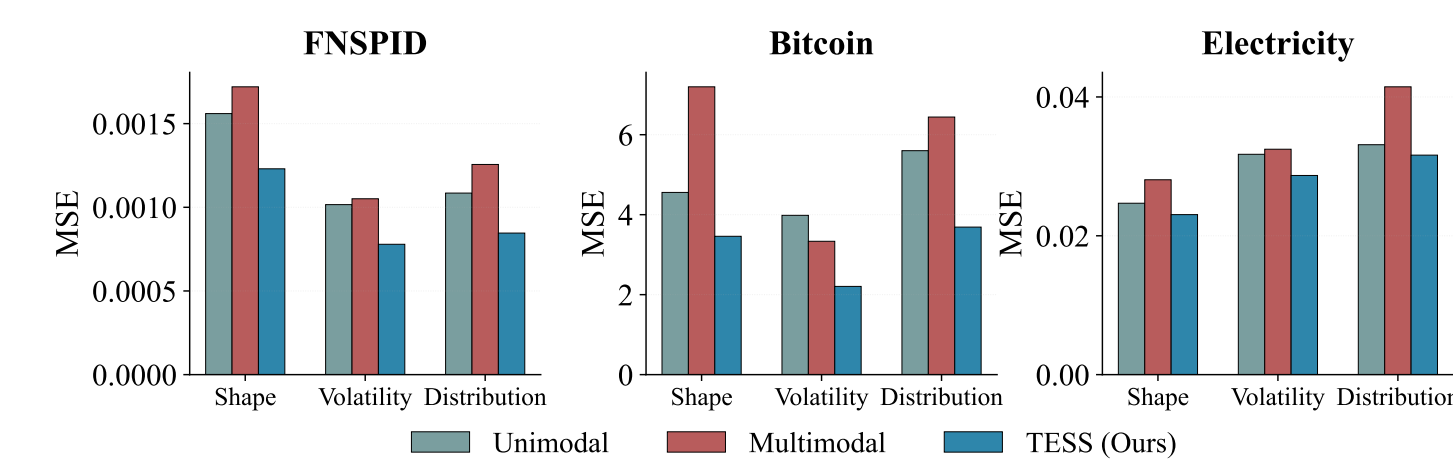
Main Results

TESS achieves the best performance on 3 of 4 datasets against 8 unimodal and 3 multimodal baselines — up to **29.1% lower MSE on Bitcoin** and **20.0% on FNSPID**.

Model	Bitcoin (Finance)			FNSPID (Finance)			Electricity (General)			Environment (General)		
	MAE	MSE	RMSE	MAE	MSE	RMSE	MAE	MSE	RMSE	MAE	MSE	RMSE
Traditional Models												
TimeMixer	1.6757	4.3725	2.0910	0.0153	0.0017	0.0407	0.1064	0.0252	0.1586	0.4219	0.3693	0.6077
TSMixer	4.1202	29.1934	5.4031	0.0353	0.0082	0.0907	0.1119	0.0258	0.1606	0.4860	0.3878	0.6227
Nonstationary	1.6003	4.3420	2.0837	0.0154	0.0016	0.0400	0.1066	0.0256	0.1600	0.4178	0.3472	0.5892
TimesNet	1.4598	3.8229	1.9552	0.0152	0.0015	0.0385	0.1035	0.0242	0.1557	0.4258	0.3473	0.5893
FEDformer	1.4197	3.2768	1.8102	0.0163	0.0016	0.0396	0.1144	0.0267	0.1633	0.4513	0.3650	0.5893
Pyraformer	1.3994	3.6377	1.9073	0.0168	0.0015	0.0383	0.1117	0.0261	0.1616	0.4303	0.3649	0.6041
Reformer	12.4630	157.8843	12.5652	0.0299	0.0141	0.1188	0.1180	0.0292	0.1709	0.4258	0.3473	0.5893
PatchTST	1.3615	3.2456	1.8016	0.0160	0.0016	0.0402	0.1047	0.0243	0.1559	0.4290	0.3706	0.6088
Multimodal Models												
TimeLLM	1.3940	3.4477	1.8568	0.0158	0.0017	0.0412	0.1074	0.0255	0.1596	0.4237	0.3714	0.6094
ChatTime	1.4774	3.9389	1.9846	0.0187	0.0018	0.0424	0.1149	0.0279	0.1670	0.4344	0.3787	0.6153
NewsForecasting	1.3593	3.2036	1.7898	0.0176	0.0017	0.0412	0.1088	0.0254	0.1593	0.4378	0.3678	0.6064
TESS (Ours)	1.1120	2.2726	1.5075	0.0147	0.0012	0.0347	0.1031	0.0230	0.1518	0.4202	0.3534	0.5945
<i>Gain vs. best (%)</i>	+18.2%	+29.1%	+15.8%	+3.3%	+20.0%	+9.9%	+0.4%	+5.0%	+2.5%	−0.6%	−1.8%	−0.9%

Comparison with baselines (lower is better). **Bold red**: best; **bold blue**: runner-up. Ablation: removing TESS raises MSE by 22.8–46.2%; removing gating by 2.6–7.5%.

Robustness to Non-Stationarity



We extract three test subsets, each representing a distinct non-stationary scenario: ① **shape transition**, ② **volatility change** and ③ **distribution shift**. TESS improves consistently across all three: MSE reductions of **21–52%** over multimodal and **21–45%** over unimodal baselines on FNSPID and Bitcoin.

References

- [1] Nie et al. A Time Series is Worth 64 Words. *ICLR*, 2023.
- [2] Kim et al. Reversible Instance Normalization (RevIN). *ICLR*, 2022.
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- [5] Wang et al. From News to Forecast. *NeurIPS*, 2024.
- [6] Liang et al. Mind the Gap. *NeurIPS*, 2022.